

UrbanCart Sales & Customer Intelligence

SQL Analysis & Excel Dashboard | January 2024–December 2025

Problem Solved

UrbanCart had two years of retail data across five disconnected files, but leadership lacked clear answers about growth, profitable products, customer retention, channel performance, discount impact, and returns. This project connects the data, answers 15 decision-focused questions with SQL, and makes the results accessible through an Excel dashboard.

Project Scope

- 3,524 orders and 6,682 order line items
- 900 customers, 180 products, and 17 stores
- 16 physical stores across four Indian regions plus one online store
- Five analytical themes: sales, products, customers, channels/regions, and discounts/returns

Headline Dashboard Results

Metric	Result
Net revenue	■10,243,322.40
Delivered orders	2,500
Average order value	■4,097.33
Repeat purchase rate	72.9%
Return rate	15.0%
Gross margin	32.6%

Figures reflect Delivered orders only and are net of discounts. Return rate is returned orders divided by all orders.

Method

- Designed a relational schema linking customers, products, stores, orders, and order items.
- Added checks for row counts, orphaned relationships, invalid quantities, prices, discounts, channels, and statuses.
- Calculated net revenue, gross profit, gross margin, average order value, and repeat purchase rate.
- Used CTEs and window functions for monthly growth, product ranking, RFM segmentation, and cohort retention.
- Built a formula-driven Excel dashboard for a non-technical audience.

Business Questions Answered

- Q1. What are the headline revenue, order, and AOV results?
- Q2. How is revenue changing month over month?
- Q3. Which months are busiest?
- Q4. Which ten products generate the most revenue?
- Q5. Which categories lead revenue, gross profit, and margin?
- Q6. What is the top product in each category?
- Q7. Who are the top 20 customers by spend?
- Q8. How do customers segment by recency, frequency, and monetary value?
- Q9. What percentage of customers purchase again?
- Q10. How does customer retention change by cohort?
- Q11. How do online and in-store performance compare?
- Q12. How does each region perform?
- Q13. Which payment methods are preferred by channel?
- Q14. Do discounts increase volume, and what do they cost in margin?
- Q15. Which categories have the highest return and cancellation rates?

Deliverables

- Five raw CSV files in the data folder
- SQL schema, loading instructions, quality checks, and 15 analytical queries
- Excel dashboard with KPI summary, live formulas, tables, and charts
- README with project context, usage instructions, metric definitions, and screenshot guidance

Metric Definitions

Net Revenue: $\text{quantity} \times \text{unit price} \times (1 - \text{discount percentage})$, for Delivered orders only.

Gross Profit: $\text{net revenue} - \text{quantity} \times \text{unit cost}$.

Average Order Value: $\text{net revenue} \div \text{delivered orders}$.

Repeat Purchase Rate: $\text{customers with more than one delivered order} \div \text{customers with a delivered order}$.

Tools

SQL (joins, CTEs, aggregations, window functions, ranking, and cohort analysis) and Microsoft Excel (formulas, KPI reporting, summary tables, and charts).